

For all following coding questions, please provide code snippets in your markdown submission file, AND take screenshots, AND submit an executable Spring project which contains your code (this project can be a clone of Springboot-Redbook, as long as it has your own code.).

1. Add newly learned annotations to your previous cheatsheet, add explanations for these annotations.
2. Walkthrough sample codes under https://github.com/CTYue/springboot-redbook/commits/06_mapper-exception, you are supposed to bring up the application on your local.
3. Explain why do we need model mappers in Spring, and in what scenarios we need it.
4. Provide 3 examples in which model mapper will NOT map successfully, explain why.
5. Explain how model mapper cast different data types between source object and target class.
6. Add your own API exceptions so that when something wrong happens in service layer, your rest API will return your customized response and status code.
7. Explain how Controller Advices work, is there any other approach to do same/similar global API exception handling?
8. What's the difference between throwing a regular exception and a customized API exception that will be eventually thrown to Controller Advice codes? Please provide screenshots to explain your findings.
9. Write some regular expression to restrict the value of attributes that your Post or Comment can have. You may use <https://regex101.com/> to construct and test/validate your regular expression.
10. Explain Spring framework fundamental principles. And how can they help build business applications?
11. Explain different types of dependency injection, explain their suitable use cases, and why field injection is not recommended in general. Please provide necessary code snippets and screenshots if possible.
12. Explain different types of application context in Spring framework, with screenshots. You may take <https://github.com/CTYue/springIOC> for reference.
13. Compare @Component and @Bean and in which scenario they should be used.
14. Explain Spring bean scopes and how to pick the correct bean scope.
15. Explain the difference between bean id and bean class.
16. Explain that when a bean has multiple alternative implementations, how will Spring decide which bean implementation to inject/autowire?